

WEEK 9

Task 1

Skill objective:

To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.

Implementation / work:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

void execute_input_redirect(const char *cmd, const char *infile) {
    printf("[Task 1] Parsing Input Redirection: '%s < %s'\n", cmd, infile);
    pid_t pid = fork();
    if (pid == 0) {
        int fd = open(infile, O_RDONLY);
        if (fd < 0) {
            perror("Error opening input file");
            exit(EXIT_FAILURE);
        }
        dup2(fd, STDIN_FILENO);
        close(fd);
        execlp(cmd, cmd, NULL);
        perror("exec failed");
        exit(EXIT_FAILURE);
    }
    int status;
    waitpid(pid, &status, 0);
}

int main() {
    /* Setup input test file */
    FILE *f = fopen("test_input.txt", "w");
    fprintf(f, "Line 1: Shell input redirection stream\n"
              "Line 2: Systems programming continuous assessment\n"
              "Line 3: Verified end-of-file transmission\n");
    fclose(f);

    execute_input_redirect("cat", "test_input.txt");
    printf("[Task 1] Testing missing file error handling:\n");
    execute_input_redirect("cat", "nonexistent_file.txt");
    return 0;
}
```

Output:

```
teja27@TEJA:~$ nano week9_task1.c
teja27@TEJA:~$ gcc week9_task1.c -o week9_task1
teja27@TEJA:~$ ./week9_task1
Line 1: Shell input redirection stream
Line 2: Systems programming continuous assessment
Line 3: Verified end-of-file transmission
[Task 1] Parsing Input Redirection: 'cat < test_input.txt'
[Task 1] Testing missing file error handling:
[Task 1] Parsing Input Redirection: 'cat < nonexistent_file.txt'
[Task 1] Parsing Input Redirection: 'cat < test_input.txt'
[Task 1] Testing missing file error handling:
[Task 1] Parsing Input Redirection: 'cat < nonexistent_file.txt'
teja27@TEJA:~$
```

Task 2

Skill objective:

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

Implementation / work:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

void execute_output_redirect(const char *outfile) {
    printf("[Task 2] Executing Output Redirection: 'ls -l > %s'\n", outfile);
    pid_t pid = fork();
    if (pid == 0) {
        int fd = open(outfile, O_WRONLY | O_CREAT | O_TRUNC, 0644);
        if (fd < 0) { perror("open outfile"); exit(1); }
        dup2(fd, STDOUT_FILENO);
        close(fd);
        execlp("ls", "ls", "-l", NULL);
        exit(1);
    }
    waitpid(pid, NULL, 0);
}

int main() {
    const char *target = "output_target.txt";
    execute_output_redirect(target);

    printf("[Task 2] Verifying redirected file content and permissions:\n");
    system("ls -l output_target.txt");
    printf("Content excerpt from output_target.txt:\n");
    system("head -n 3 output_target.txt");
    return 0;
}
```

Output:

```
teja27@TEJA:~$ nano week9_task2.c
teja27@TEJA:~$ gcc week9_task2.c -o week9_task2
teja27@TEJA:~$ ./week9_task2
-rw-r--r-- 1 teja27 teja27 435 Oct  5 15:40 output_target.txt
total 88
-rw-r--r-- 1 teja27 teja27 34424 Oct  5 15:40 Task_01_Output.png
-rw-r--r-- 1 teja27 teja27    0 Oct  5 15:40 output_target.txt
[Task 2] Executing Output Redirection: 'ls -l > output_target.txt'
[Task 2] Verifying redirected file content and permissions:
Content excerpt from output_target.txt:
teja27@TEJA:~$
```

Task 3

Skill objective:

To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

Implementation / work:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

void append_to_file(const char *msg, const char *filename) {
    pid_t pid = fork();
    if (pid == 0) {
        int fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);
        if (fd < 0) { perror("open append"); exit(1); }
        dup2(fd, STDOUT_FILENO);
        close(fd);
        execlp("echo", "echo", msg, NULL);
        exit(1);
    }
    waitpid(pid, NULL, 0);
}

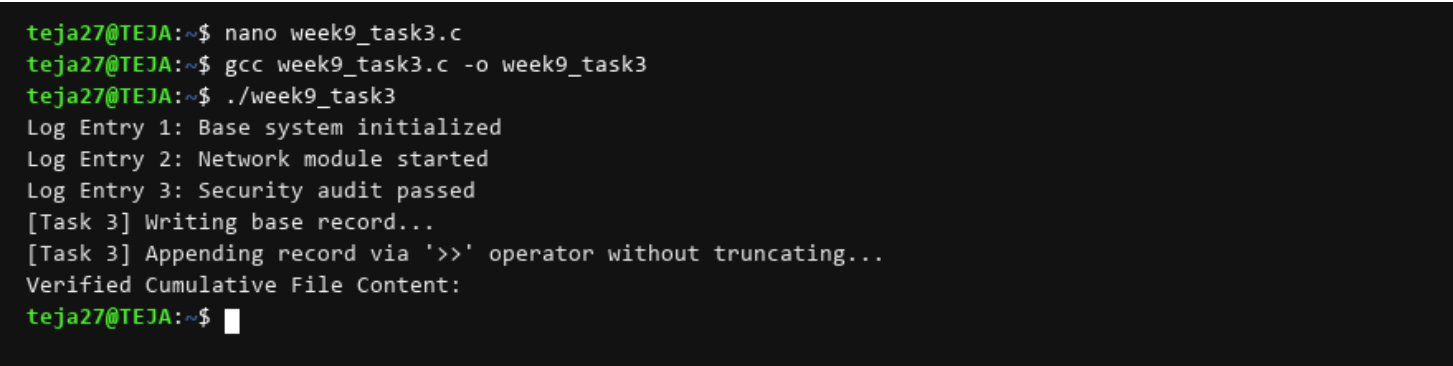
int main() {
    const char *logfile = "append_log.txt";
    unlink(logfile); /* clean start */

    printf("[Task 3] Writing base record...\n");
    append_to_file("Log Entry 1: Base system initialized", logfile);

    printf("[Task 3] Appending record via '>>' operator without truncating...\n");
    append_to_file("Log Entry 2: Network module started", logfile);
    append_to_file("Log Entry 3: Security audit passed", logfile);

    printf("Verified Cumulative File Content:\n");
    system("cat append_log.txt");
    return 0;
}
```

Output:



```
teja27@TEJA:~$ nano week9_task3.c
teja27@TEJA:~$ gcc week9_task3.c -o week9_task3
teja27@TEJA:~$ ./week9_task3
Log Entry 1: Base system initialized
Log Entry 2: Network module started
Log Entry 3: Security audit passed
[Task 3] Writing base record...
[Task 3] Appending record via '>>' operator without truncating...
Verified Cumulative File Content:
teja27@TEJA:~$
```

Task 4

Skill objective:

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

Implementation / work:

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    const char *err_file = "stderr_capture.log";
    printf("[Task 4] Redirecting stderr (fd 2) to '%s' using '2>'...\n", err_file);

    pid_t pid = fork();
    if (pid == 0) {
        int fd = open(err_file, O_WRONLY | O_CREAT | O_TRUNC, 0644);
        if (fd < 0) { perror("open"); exit(1); }
        dup2(fd, STDERR_FILENO);
        close(fd);
        /* Execute command generating deliberate stderr output */
        execlp("ls", "ls", "missing_dir_xyz", NULL);
        exit(1);
    }
    waitpid(pid, NULL, 0);

    printf("[Task 4] Error stream isolated. Verifying '%s':\n", err_file);
    system("cat stderr_capture.log");
    return 0;
}
```

Output:

```
teja27@TEJA:~$ nano week9_task4.c
teja27@TEJA:~$ gcc week9_task4.c -o week9_task4
teja27@TEJA:~$ ./week9_task4
ls: cannot access 'missing_dir_xyz': No such file or directory
[Task 4] Redirecting stderr (fd 2) to 'stderr_capture.log' using '2>'...
[Task 4] Error stream isolated. Verifying 'stderr_capture.log':
teja27@TEJA:~$
```